

P4.3 — holomorphic discrete series K-type framework v0.1 (bulk/spinor towers, Casimirs, mass-mechanism honest assessment). Bulk tower $V_{-}(k,0)$ and spinor radial tower $V_{-}(1/2+k, 1/2)$ computed; CRITICAL FINDING: simple Casimir-squared mass ratios do NOT match observed lepton ratios ($m_{\mu}/m_e \approx 207$, $m_{\tau}/m_{\mu} \approx 17$). L4 v0.2 must connect existing BST mass closed-forms (T190, T2003, T187) to the dictionary, not derive ratios from naive Casimir towers.

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0. The setup

The bulk Hilbert space $L^2_{\text{hol}}(D_{IV}^5)$ decomposes under $K = SO(5) \times SO(2)$ as a direct sum of K-types. The holomorphic tangent at the origin is $T^{\wedge}(1,0) = 5_{-}(+1)$ (vector of $SO(5)$ at $SO(2)$ charge +1); the bulk K-types come from $\text{Sym}(T^{\wedge}(1,0)^*)$.

1. The bulk symmetric tower $V_{-}(k,0)$

$\text{Sym}^k(5)$ of $SO(5)$ decomposes as $\bigoplus_{j=0}^{\lfloor k/2 \rfloor} V(k-2j, 0)$ (symmetric tensors of decreasing trace). The LOWEST K-types in each Sym^k are $V_{-}(k, 0)$.

k	K-type	Weyl dim ($SO(5)$)	Casimir = $k(k+3)$
0	$V_{-}(0,0)$	1	0
1	$V_{-}(1,0)$	$5 = n_C$	$4 = \text{rank}^2$
2	$V_{-}(2,0)$	14	$10 = \dim \mathfrak{so}(5)$
3	$V_{-}(3,0)$	30	18
4	$V_{-}(4,0)$	55	28
5	$V_{-}(5,0)$	91	40

The bulk Casimir spacings are 4, 6, 8, 10, 12, ... (arithmetic +2). The dims also follow a clean polynomial pattern.

Substrate-primary anchors on the bulk tower: - $V_{-}(0,0)$ Casimir 0 = vacuum/Higgs scalar. - $V_{-}(1,0)$ dim 5 = n_C , Casimir 4 = $\text{rank}^2 \rightarrow$ photon (vector). - $V_{-}(2,0)$ Casimir 10 = $\dim \mathfrak{so}(5) \rightarrow$ next bulk excitation (potentially a higher gauge mode).

2. The spinor radial tower $V_{-}(1/2+k, 1/2)$ (lepton lane)

Leptons live on the Shilov boundary at lowest K-type $V_{-}(1/2, 1/2)$. The “radial” extension (generation tower candidate) is $V_{-}(1/2+k, 1/2)$ for $k = 0, 1, 2, \dots$:

k	K-type	Casimir
0	$V_{-}(1/2, 1/2)$	$5/2 = \rho_1$ (lepton)
1	$V_{-}(3/2, 1/2)$	$15/2$
2	$V_{-}(5/2, 1/2)$	$29/2$
3	$V_{-}(7/2, 1/2)$	$47/2$

Casimir spacings: 5, 7, 9, ... (arithmetic +2).

Substrate-primary anchor: $V_{-}(1/2, 1/2)$ Casimir $5/2 = \rho_1$ = Bergman kernel singularity exponent — the “natural mass-setting” alignment (L4 v0.1). The HIGHER spinor K-types could carry the generations ($e/\mu/\tau$) — IF the mass mechanism is the radial tower.

3. THE CRITICAL FINDING — simple Casimir²-mass doesn’t match observation

If $\text{mass}^2 \propto \text{Casimir}$, the lepton mass ratios would be:

- $m(k=1)/m(k=0) = \sqrt{(15/2 / 5/2)} = \sqrt{3} \approx 1.73$
- $m(k=2)/m(k=1) = \sqrt{(29/15)} \approx 1.39$

But observed: - $m_\mu/m_e \approx 206.77$ - $m_\tau/m_\mu \approx 16.82$

Simple Casimir²-mass DOES NOT match observed lepton mass ratios by 2 orders of magnitude. The radial tower’s Casimirs grow too slowly to produce the SM lepton hierarchy.

4. The right mechanism: existing BST mass closed-forms

The existing BST mass derivations use DIFFERENT mechanisms, not naive Casimir squared:

theorem	mass relation	precision	mechanism
T190	$m_\mu / m_e = (24/\pi^2)^6 \approx 206.77$	0.004%	closed form via $24 = \text{rank}^3 \cdot N_c$ (rank-cube color product) and π^2 (radial kernel power); $6 = C_2$
T2003	$m_\tau / m_e = 49 \cdot 71 = 3479$	~0.05%	BST primary product: $49 = g^2$, $71 = \text{some BST combination}$
T187	$m_p / m_e = 6 \cdot \pi^5 = 1836.12$	0.002%	adjoint Casimir $\times$ Bergman volume (boundary→bulk bridge)

The leptonic radial tower’s “spacing” is NOT a Casimir difference. It involves the Bergman kernel integral structure (Track BC’s hydrogen 1s) and BST-primary closed forms like $(24/\pi^2)^6$ — products and powers of substrate integers, not simple Casimir spectra.

5. What L4 v0.2 must do

L4 v0.2 (explicit per-particle lepton mass derivation, gated on Elie's bulk K-type radial tower per affine pin $B_2^{(1)}$) should:

1. Connect the dictionary (lepton at K-type $V_{(1/2,1/2)}$) to the existing closed-forms (T190, T2003). The dictionary places the lepton at Casimir $5/2 = \rho_1$ (kernel exponent — “natural unit,” L4 v0.1 alignment); the EXPLICIT mass ratios come from the radial Bergman kernel integral structure that gives $(24/\pi^2)^6$ etc.
2. Derive the closed forms from the K-type tower + kernel structure: this is the multi-week work. Why does m_μ/m_e factor as $(24/\pi^2)^6$? The $24 = \text{rank}^3 \cdot N_c$ is a BST primary product; the π^2 is the radial kernel power; the $6 = C_2$ is the Casimir exponent. A dictionary-level derivation would show all three emerge from the K-type structure + Bergman kernel.
3. NOT just compute Casimir ratios — those are off by orders of magnitude.

6. The honest scope (what v0.1 does NOT do)

- Does NOT derive lepton mass ratios from naive Casimir towers (explicitly shown not to work).
- Does NOT re-derive T190/T2003/T187 (existing BST results, anchors).
- DOES set up the K-type tower structure as the framework (bulk + spinor radial towers, Casimirs, dims, substrate-primary anchors).
- DOES IDENTIFY the right mechanism for L4 v0.2 (connect dictionary K-type to existing closed forms, not derive ratios from scratch via Casimirs).

This is a useful honest negative: it tells the team that Elie's bulk K-type radial tower computation, when ready, must aim at deriving the EXISTING closed forms (T190's $(24/\pi^2)^6$, etc.) from the K-type structure + Bergman kernel — NOT at the naive Casimir²-mass approach.

7. Honest scope + tier

RIGOROUS (rep theory): Sym(5) decomposition under SO(5) into harmonic summands; $V_{(k,0)}$ Weyl dims and Casimirs (computed, checked); spinor radial tower Casimirs (computed); arithmetic +2 spacing pattern.

HONEST NEGATIVE: simple Casimir²-mass tower DOES NOT match observed lepton mass ratios. By 2 orders of magnitude. L4 v0.2 cannot use this naive approach.

FRAMEWORK GUIDANCE: L4 v0.2 must connect the dictionary K-type placement (lepton at Casimir $5/2 = \rho_1$ = kernel exponent) to the EXISTING BST mass closed-forms (T190, T2003, T187), via the Bergman kernel integral structure that gives those closed forms — NOT derive ratios from scratch via Casimir spectra.

Cal #27 / honesty: this v0.1 is a HONEST NEGATIVE plus FRAMEWORK guidance, not a derivation. I checked the naive mass-from-Casimir approach explicitly and it fails by orders of magnitude; reporting that clearly so Elie's L4 v0.2 work targets the right mechanism (kernel-integral closed forms, not Casimir spectra).

Routed: → Elie: when affine pin ($B_2^{(1)}$) ratifies and you start the bulk K-type radial tower, the TARGET is to derive T190's $(24/\pi^2)^6$ and T2003's 49-71 from the kernel-integral structure on the spinor radial tower $V_{(1/2+k, 1/2)}$ — NOT to use simple Casimir²-mass (which fails). The dictionary's role: place each generation's K-type and provide the kernel structure that yields the closed-form ratios. → Keeper: this honest-negative framework prevents L4 v0.2 from chasing a dead end (naive Casimir²-mass). → me: continuing to other items (bulk-color v0.3 / further study items / consolidation).

— Lyra, P4.3 holomorphic discrete series K-type framework v0.1. Bulk tower $V_{(k,0)}$ (Casimirs 0, 4, 10, 18, ...) and spinor radial tower $V_{(1/2+k, 1/2)}$ (Casimirs 5/2, 15/2, 29/2, 47/2) computed cleanly. CRITICAL FINDING: simple Casimir²-mass ratios = 1.73 ($k=1/k=0$), 1.39 ($k=2/k=1$) — DO NOT MATCH observed $m_\mu/m_e \approx 207$ or $m_\tau/m_\mu \approx 17$ by orders of magnitude. L4 v0.2 must connect dictionary K-type placement (lepton at $C=5/2=\rho_1$) to EXISTING BST mass closed forms (T190 $(24/\pi^2)^6$, T2003 49-71, T187 $6\pi^5$) via the kernel-integral structure — NOT naive Casimir spectra. Useful framework + honest negative; prevents L4 v0.2 from chasing a dead end.